

National Exam of Higher National Diploma-New program – 2020 Session

Spécialty/option : Software Engineering
Paper : System Analysis and Design
Duration : 4 hours



Credit : 7

SECTION A:

INFORMATION SYSTEMS AND DATABASE (50 marks).

I: OBJECT MODELING (25 marks).

Exercise I

MCQ'S each question carries 1 mark (10 marks)

1. The detailed specification of how all the parts of a system will be implemented and coordinated is called.

- A) programming B) paraphrasing C) system design D) structuring

2. The primary purpose of a good modeling techniques is to _____

- A) to promote communication B) increase functional cohesion
C) reduce the need for structure D) reduce dependency between modules

3. The UML provides standard ways to do all of the following to business systems except them.

- A) construct B) document C) describe D) destroy

4. The UML is commonly used to model all of the following except.

- A) computer programs B) business activities
C) organizational processes D) software systems

5. The UML was intentionally designed to be _____

- A) low-level, detail-oriented B) used with Visual Basic
C) nontechnical D) inexpensive

6. The UML diagrams that show how a business works from the perspective of those who actually interact with the business, such as employees or customers, are diagrams.

- A) communication B) use case C) state machine D) class

7. Which of the following would be portrayed as an extend relationship in a use case diagram for a hospital?

- A) the relationship between the head nurse and the floor nurses
B) admitting a patient who has never been admitted before.
C) serving a meal.
D) scheduling the monitoring of patients' vital signs

8. The people shown in use case diagrams are called _____

- A) workers B) clowns C) actors D) relatives

9. One aspect of use case diagrams that makes them difficult to learn is that _____

- A) they require programming experience to understand

- B) they use a technical vocabulary
- C) there is no single right answer for any case.
- D) all of the above

10. The arithmetic association relationship between a college student and college courses would be expressed as _____

- A) 1 0
- B) 1 1
- C) 1 0...*
- D) 0...* 0

Exercise 2 (15 MARKS)

1. a) What is object? (2 marks)
b) Discuss the main characteristics of the object with example from the real world. (4marks)
2. What is: (4 marks)
a) UML?
b) Modeling
3. Explain two types of system development process of your choice. (3marks)
4. Differentiate between aggregation and composition. (2marks)

DATABASE ADMINISTRATION WITH MYSQL (25 marks).

Draw an entity-relationship diagram for the following information system. The Aeroporter Transport Services Company runs a limousine service that carries passengers to and from the Toronto airport to their homes or places of business. They maintain a database of customers on a PC in order to schedule pickups and also to keep their customers from having to repeat address information each time they call the limousine service. The database for the customers is accessed by customer telephone number. If a customer is picked up sometimes at their home and sometimes at their office, both home and office telephone numbers and addresses are stored in the database. The customer may also schedule a pickup from the airport when his or her flight arrives or may call from the airport and reserve a limousine which will come in approximately five minutes. When a request for a pickup comes in, the dispatcher checks for available drivers, calls one and assigns them to a customer. Typically cars are assigned to the driver each workday and often a driver will take a car home for an early morning pickup if needed.

The relevant customer and driver attributes kept in the database for managing the customer pickup and delivery is as follows:

- Customer Name
- Customer Home Telephone Number
- Customer Home Address
- Customer Home Region
- Customer Work Telephone Number
- Customer Work Address
- Customer Work City Region Date of Pickup

1. Time of Pickup Driver assigned to Pickup Special Pickup Information Flight Number Air Carrier Arrival Time Customer Drop-off City Region Driver ID Driver Name Driver Address Driver City Region Driver Phone Number Car ID Car Make Car License Number

QUESTION 1. WEB PROGRAMING (25MARKS)

1. Give the full meaning of the following Acronyms (8mks)

- a. HTML _____
- b. W3C world wide web consortuim
- c. CSS _____
- d. HTTP hypertext transfer protocol
- e. IP _____
- f. URL uniform resource locator
- g. DNS _____
- h. FTP _____

2. Match the technology or practice on the left with the problem it best addresses: (5 Marks)

Technology	Problem
1. ___ Progressive enhancement	a. Assistive reading and input devices
2. ___ Server-side detection	b. Slow connection speeds
3. ___ Responsive design	c. All levels of browser capabilities
4. ___ WAI-ARIA	d. Determining which device is being used
5. ___ Site performance optimization	e. A variety of screen sizes

3. Give the function of each of the following programming languages: HTML, CSS, JavaScript, PHP. (5marks)
4. List two software you can use to type a HTML program (2 marks)
5. What type of storage engine MySQL support (5 marks)

QUESTION 2. MOBILE PROGRAMING (25MARKS)

1. What is mobile programming? (2marks)
2. What is an android software? (2marks)
3. What is a mobile terminals? (2marks)
4. State the different types of mobile terminals that you know (3marks)
5. What is an android operating system? (2marks)
6. State two examples of android operating system (2marks)
7. What is a web application? (2marks)

8. Protecting the Security and Privacy of Mobile Devices (10 marks)

As the number of smartphone and tablet devices increases, so does the risk that hackers and computer criminals will target these devices. Since they were first introduced, certain

mobile devices have been vulnerable to eavesdropping and fraudulent charges. In addition, many of them automatically track a user's location.

The major security risks associated with mobile devices include: *Malware*—Android apps, in particular, are vulnerable to malware because of the platform's openness.

To guard against malware threats, users have to upload the most recent versions of the operating system and use mobile security tools. *Premium SMS billing*—With their devices' vulnerability to malware, smartphone users face an added risk of subscribing to premium text-messaging services that charge every time users interact with them.

Most cell phone carriers allow subscribers to block premium SMS messaging,

however. *E-mail and SMS phishing*—Because it is more difficult to establish a link's legitimacy on a mobile device, mobile users are more likely to click on them, which is a phisher's dream come true. Mobile users should therefore use a lot of caution when using e-mail on these devices. *Spyware*—Commercially available software can be used by intruders to track and control the user's mobile activities.

Malicious Web sites—These could pose a threat in the future, given that many smartphone browsers are based on a browser engine with vulnerabilities. Never leave your mobile device unlocked. Never leave it unattended. Always protect it with a password.

Answer the following questions:

- a) What are some examples of security risks associated with a mobile device?
- b) How can these devices automatically track a user's location?
- c) What are a couple of recommendations for protecting your mobile device against these threats?